

Name: _____

Registration: _____

Design an 8x1 Mux using as building blocks the 2x1 and 4x1 Muxes shown below. Use the provided skeleton file to implement the 8x1 Mux.

(Simulation: 50% pts., FPGA Implementation: 50% pts.)

```
module m21 (OUT, A, B, S);  
    input A, B, S;  
    output OUT;  
  
    assign OUT=S?B:A;  
endmodule
```

```
module m41 (OUT, A, B, C, D, S1, S0);  
    input A, B, C, D, S1, S0;  
    output OUT;  
  
    always @(*)  
        case ({S1, S0})  
            0: OUT=A;  
            1: OUT=B;  
            2: OUT=C;  
            3: OUT=D;  
        endcase  
endmodule
```

```
module m81 (OUT, I, Sel);  
    input [7:0] I;  
    input [2:0] Sel;  
    output OUT;  
    //Write your code below  
  
endmodule
```